

Underwater Robot to Clean Tanks

SaUsR underwater robot developed by Gridbots can clean water tanks without the need for draining the water tank completely



SaUsR robot

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Fantasies depicted in TV shows such as *The Jetsons* and *Futurama* are increasingly getting closer to reality. A step in this direction is the robot created by Gridbots to clean water tanks while these are still filled with water.

Gridbots was established in 2007 in Rajasthan with an aim to develop robots that can solve real-world problems. Pulkit Gaur, founder and chief technology officer, observed that water was scarce in Rajasthan and Gujarat, and cleaning water tanks was not a common practice as the only way to clean a tank was to empty it and clean manually, causing wastage of water.

To avoid wastage of water, people filtered water through a multi-layered

cloth instead, which did not help much in removing germs that are present at the bottom of water tanks.

Gaur and his team started working on a prototype that could clean water tanks without wasting water. Within three months, Gaur's team created a new mechanism for cleaning water storage tanks that did not cause downtime or wastage of water. They named it SaUsR, short for 'smart autonomous underwater service robot.' SaUsR's low power consumption, small size and light weight set it apart from other tank cleaning devices. Cleaning of the tank while water is still in it reduced water wastage to 5-10 per cent of the tank water.

"We have devised the complete system after a long research on water tanks, especially those found in Rajas-

than and Maharashtra," shares Gaur.

The robot weighs 30 kilograms and works off a single-phase power supply. It is 61 cm (2 feet) wide and as much long, which ensures that it can easily get into Indian water tanks. A 91×61 cm² model is being worked upon for industrial purposes. A 50-metre cable is attached to the robot. Hence the robot can cover as much depth with a linear speed of five centimetres per second and underwater payload (carrying capacity) of 30 kilograms.

SaUsR consumes 500-800 watts of power based on load conditions. Shelf life is estimated to be five years. SaUsR cannot be purchased yet, but it can be leased from Gridbots for Rs 1.5 million a year. Most of this amount is refundable.

How it works

SaUsR has a specially designed waterproof locomotion system to work underwater. It uses a simple cleaning and suction mechanism that is particularly suited for cleaning the sludge found in Indian tanks. Gridbots developed waterproof drive electronic components that let the user control the motor and navigate it using a laptop. The team used parts such as three-phase motors, heavy suction systems and pneumatic systems. They also developed a waterproof camera with a pan-tilt mechanism to provide a video feed of the tank.

The machine consists of three units—the waterproof robot unit that can move underwater and be controlled with a remote, a pumping unit and a counter suction unit with a cavity. The robot inserted into the tank uses visual sensors with high-resolution cameras mounted on its back to provide live video for better control. It uses light-emitting diodes for lighting